OSPF Link State Routing

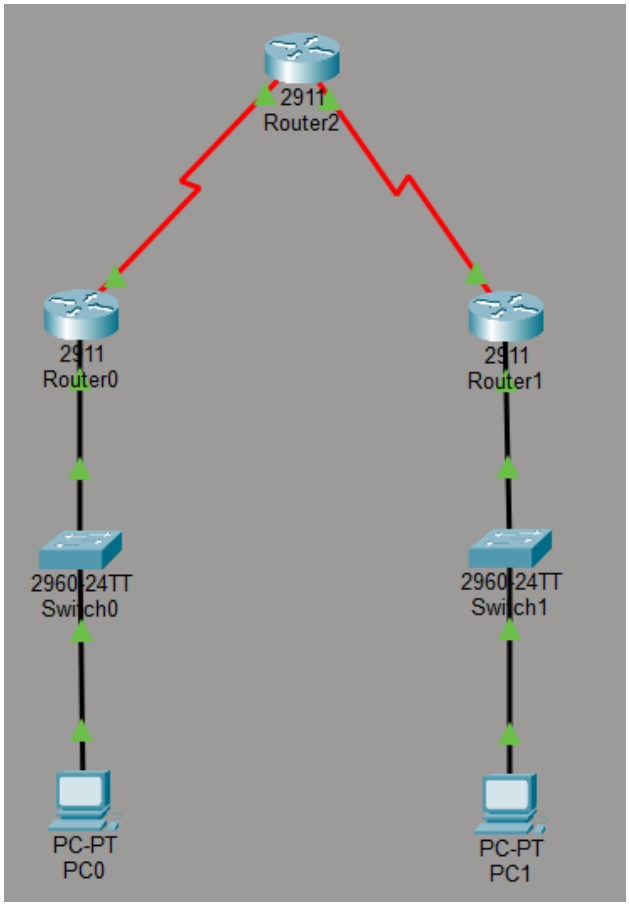
Assignment 7

Objective

To configure OSPF (Open Shortest Path First) routing on a 3-router topology, analyze the Link State Database, and verify route selection using SPF algorithm and OSPF cost.

Screenshots & Observations

Screenshot 1: Physical Topology — All Green



Screenshot 2: OSPF Neighbor Table — FULL state

```
Router>enable
Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           0   FULL/  -        00:00:39    100.1.1.1       Serial0/3/0
3.3.3.3           0   FULL/  -        00:00:31    20.1.1.2        Serial0/3/1
Router#
```

Screenshot 3: Routing Table

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 1 subnets
O       10.2.2.0/24 [110/65] via 20.1.1.2, 00:43:29, Serial0/3/1
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.1.1.0/24 is directly connected, Serial0/3/1
L       20.1.1.1/32 is directly connected, Serial0/3/1
     100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.1.1.0/24 is directly connected, Serial0/3/0
L       100.1.1.2/32 is directly connected, Serial0/3/0
O    172.16.0.0/16 [110/65] via 100.1.1.1, 00:43:29, Serial0/3/0
```

Screenshot 4: OSPF Link State Database

```
Router#show ip ospf database
            OSPF Router with ID (1.1.1.1) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
1.1.1.1         1.1.1.1         833         0x80000006 0x00d931 4
3.3.3.3         3.3.3.3         834         0x80000005 0x008ace 3
2.2.2.2         2.2.2.2         833         0x80000005 0x007e94 3
Router#
```

Screenshot 5: PC0 Ping to PC1

```
C:\>ping 10.2.2.20

Pinging 10.2.2.20 with 32 bytes of data:

Request timed out.
Reply from 10.2.2.20: bytes=32 time=5ms TTL=125
Reply from 10.2.2.20: bytes=32 time=9ms TTL=125
Reply from 10.2.2.20: bytes=32 time=2ms TTL=125

Ping statistics for 10.2.2.20:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 9ms, Average = 5ms
```

Screenshot 6: Traceroute PC0 to PC1

```
C:\>tracert 10.2.2.20

Tracing route to 10.2.2.20 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      172.16.2.1
  2   1 ms      9 ms      0 ms      100.1.1.2
  3   2 ms      11 ms     1 ms      20.1.1.2
  4   0 ms      0 ms      0 ms      10.2.2.20

Trace complete.
```